Lecture 4

Symmetries are unitary or antiunitary

Objectives. *A bijection of all rays preserving transition probabilities comes from a unitary or an antiunitary operator. Its type is global, and its only remaining freedom is a phase. Complex conjugation supplies time reversal on the line; on a flux-threaded ring it reverses both momentum and flux. Every member of a one-parameter group of ray symmetries has unitary type.*

- Prove Wigner's theorem, including one global linear or antilinear choice and uniqueness up to phase.
- Compute time reversal on the line and a twisted ring, keeping operator domains and flux signs.
- Derive the odd ground-state current away from crossings, and explain why an antiunitary cannot be a member of a real one-parameter symmetry group.

4.1 What must a symmetry preserve?

QM-I called a unitary commuting with the Hamiltonian a symmetry (Definition 6.3). It also introduced an antiunitary time reversal (Definition 7.6). The two notions were not the same: time reversal is antilinear, so it was not a symmetry in the sense of Definition 6.3. Here we ask the kinematic question before specifying a Hamiltonian. Which changes of pure states preserve the probabilities for comparing them?

Definition 4.1 (Ray symmetry)**.** The *ray* $[\psi]$ of a nonzero vector is its one-dimensional complex span. For two rays their *transition probability* is

$$p([\psi],[\varphi]) = \frac{|\langle\psi,\varphi\rangle|^2}{\|\psi\|^2\,\|\varphi\|^2}.$$

A *ray symmetry* (or Wigner symmetry) is a bijection $\mathcal{S}$ of all rays preserving these transition probabilities.

Rescaling either vector cancels between numerator and denominator. For unit vectors the probability is the expectation of $|\varphi\rangle\langle\varphi|$ in $|\psi\rangle\langle\psi|$, as in QM-I and L2. Orthogonal rays have transition probability zero; identical rays have probability one. Both relations are therefore preserved by a symmetry and its inverse. The definition is about all rays of the specified space. Restrictions to coherent sectors will be stated separately in Lecture 5.

Definition 4.2 (Antiunitary)**.** An *antiunitary* $\mathcal{V} : \mathcal{H} \to \mathcal{H}$ is an onto antilinear map such that

$$\mathcal{V}(a\psi + b\varphi) = \bar{a}\mathcal{V}\psi + \bar{b}\mathcal{V}\varphi, \qquad \langle \mathcal{V}\psi, \mathcal{V}\varphi \rangle = \overline{\langle \psi, \varphi \rangle}.$$

A unitary is onto, linear and preserves the inner product itself.

Both kinds preserve norms and transition probabilities, and hence induce Wigner symmetries. Multiplying an implementation by a complex number of modulus one changes no image ray. It remains to prove the converse and to rule out a map that makes an independent linear/antilinear choice for each vector. That last consistency question is the main work.

Self-check 4.1

1. Why is $\psi \mapsto \bar{\psi}$ on $\mathbb{C}^2$ antilinear rather than linear?
2. Does every Wigner symmetry preserve a given Hamiltonian? Which assumption would add that requirement?

4.2 Wigner's theorem: fixing an anchor ray

Theorem 4.3 (Wigner)**.** *On a separable complex Hilbert space with* $\dim \mathcal{H} \geq 2$*, every Wigner symmetry is induced by a unitary or antiunitary* $\mathcal{V}$*, meaning* $\mathcal{S}[\psi] = [\mathcal{V}\psi]$ *for every nonzero* ψ*. Its type is unique, and any two implementations differ by one complex phase.*

The proof has one organising idea: fix an output phase by comparison with an anchor, recover real inner products, and only then determine the complex type. Absolute overlaps alone do not fix arbitrary vector phases. The anchored superpositions supply the missing relative information.

Proof. 1. Choose an anchor and fix phases relative to it. Choose a unit vector e and a unit representative f of $\mathcal{S}[e]$. Put $K = e^{\perp}$ and $K' = f^{\perp}$. For nonzero $z \in K$, choose a unit representative g of $\mathcal{S}[z]$. Then $g \perp f$. The squared projections of $(e+z)/\sqrt{1+\|z\|^2}$ onto e and $z/\|z\|$ sum to one. The image ray has the same two probabilities with f, g, so a unit representative has no orthogonal remainder: Pythagoras puts it in $\operatorname{span}\{f, g\}$. Its f component is nonzero. Rescale uniquely to make that component equal to one, writing

$$\mathcal{S}[e+z] = [f + Wz], \qquad Wz \in \mathcal{S}[z], \quad Wz \perp f. \tag{4.1}$$

The transition with $[e]$ gives $1/(1+\|Wz\|^2) = 1/(1+\|z\|^2)$, so $\|Wz\| = \|z\|$. Put $W0 = 0$. This defines $W : K \to K'$ before any linearity assertion; the fixed anchor coefficient removes the otherwise arbitrary output phase.

2. Recover real inner products. For $z, w \in K$, transition preservation for their rays and their anchored rays gives

$$|\langle Wz, Ww \rangle| = |\langle z, w \rangle|, \qquad |1 + \langle Wz, Ww \rangle| = |1 + \langle z, w \rangle|.$$

The zero-vector cases hold directly. The normalising denominators cancel because norms were already shown equal. Subtract the squared equalities, using $|1+c|^2 = 1 + 2\operatorname{Re} c + |c|^2$, to obtain

$$\operatorname{Re}\langle Wz, Ww \rangle = \operatorname{Re}\langle z, w \rangle. \tag{4.2}$$

3. Real linearity follows from these Gram entries. Expand $\|W(z+w) - Wz - Ww\|^2$. Each norm and each real inner product in the expansion equals its input counterpart by (4.2). Thus the expansion is

$$\|z+w\|^2 + \|z\|^2 + \|w\|^2 - 2\,\mathrm{Re}\,\langle z+w, z\rangle - 2\,\mathrm{Re}\,\langle z+w, w\rangle + 2\,\mathrm{Re}\,\langle z, w\rangle = 0.$$

This is $\|(z+w) - z - w\|^2$, giving additivity. Equation (4.2) applies to every pair of inputs, including $z+w, z$ and $z+w, w$. For real r, the same argument applied to $\|W(rz) - rWz\|^2$ gives $W(rz) = rWz$. Explicitly, its terms are $r^2\|z\|^2 + r^2\|z\|^2 - 2r\,\mathrm{Re}\,\langle rz, z\rangle = 0$. In particular W is injective, since it preserves norms.

4. There is one global complex type. Since iz and z have the same ray, $W(iz) = \lambda Wz$ for some $|\lambda| = 1$ when $z \neq 0$. Equation (4.2) gives $\mathrm{Re}\,\langle Wz, W(iz)\rangle = \mathrm{Re}\,\langle z, iz\rangle = 0$, so $\mathrm{Re}\,\lambda = 0$. Consequently $W(iz) = \sigma(z)iWz$, with $\sigma(z)$ either $+1$ or -1.

If z, w are real-linearly independent, real additivity gives

$$\sigma(z+w)i(Wz+Ww) = W(i(z+w)) = \sigma(z)iWz + \sigma(w)iWw.$$

The vectors iWz, iWw are real-linearly independent: a real relation, after multiplication by $-i$, would contradict real linearity and injectivity of W. Comparing their coefficients therefore gives $\sigma(z) = \sigma(w) = \sigma(z+w)$. If instead $w = rz$ for nonzero real r, real homogeneity applied to $iw = r(iz)$ directly gives $\sigma(w) = \sigma(z)$. These cases cover every pair of nonzero vectors. Hence one sign works throughout K, even when K is one complex dimension.

In the plus case W is complex linear, since $W(iz) = iWz$; in the minus case it is antilinear. Combining real-inner-product preservation with $\mathrm{Re}\,\langle iz, w\rangle = \mathrm{Im}\,\langle z, w\rangle$ shows that W preserves the full inner product in the first case and conjugates it in the second.

5. Restore the anchor and all rays. Write $\chi(a) = a$ in the linear case, $\chi(a) = \bar{a}$ in the antilinear case, and define

$$\mathcal{V}(ae+z) = \chi(a)f + Wz \qquad (z \in K).$$

Orthogonality to the anchors proves the same norm and inner-product properties on all of $\mathcal{H}$. For $a = 0$ the image-ray statement is already in the construction. For $a \neq 0$, use $[ae+z] = [e + a^{-1}z]$ in (4.1) and multiply the output representative by $\chi(a)$. Homogeneity of W then gives $\mathcal{S}[ae+z] = [\mathcal{V}(ae+z)]$. Its range is a complex subspace meeting every ray, by surjectivity of $\mathcal{S}$. It is therefore all of $\mathcal{H}$, making $\mathcal{V}$ unitary or antiunitary as claimed. This also covers vectors orthogonal to the anchor.

6. Prove uniqueness, including the type. If two implementations have the same type, their quotient is linear and fixes every ray. For independent x, y, write its actions as ax, by; fixing $[x+y]$ forces $a = b$. Collinear vectors have the same scalar by linearity, and comparison with an independent vector connects every ray. Thus the quotient is one scalar of modulus one. If the types differed, their quotient would be an antilinear map fixing all rays. On orthonormal e_1, e_2, fixing $[e_1 + e_2]$ forces equal scalars on the two basis vectors. Antilinearity then maps $[e_1 + ie_2]$ to $[e_1 - ie_2]$, a different ray. This contradiction establishes type uniqueness for dimension at least two. ∎

The theorem is kinematic: it does not say that every such map commutes with the energy or is an available time evolution.

Example 4.4 (Why the complex type cannot vary with the vector)**.** On $\mathbb{C}^2$ take anchor $e = f = |0\rangle$ and write $z = \zeta\,|1\rangle$. Conjugation gives $W(\zeta\,|1\rangle) = \bar{\zeta}\,|1\rangle$ throughout the orthogonal line. A diagonal unitary $\mathrm{diag}(1, e^{i\beta})$ instead gives $W(\zeta\,|1\rangle) = e^{i\beta}\zeta\,|1\rangle$ throughout it. These are two

consistent choices; the anchor proof does not choose the image of each individual coefficient independently.

To see what can go wrong, try using $W\zeta = \zeta$ when $\operatorname{Re}\zeta \geq 0$ but $W\zeta = \bar{\zeta}$ otherwise. For $z = (1+i)\,|1\rangle$ and $w = (-1+i)\,|1\rangle$, their inner product is $2i$. The attempted images have inner product -2. The anchored vectors all have norm $\sqrt{3}$, but their transition probabilities would be

$$p([e+z],[e+w]) = \frac{|1+2i|^2}{9} = \frac{5}{9}, \qquad p([f+Wz],[f+Ww]) = \frac{|1-2|^2}{9} = \frac{1}{9}.$$

Thus the attempt already fails on ray probabilities, before questions about surjectivity arise. Rays lying in $e^{\perp}$ alone could not detect the error: that complex line has only one ray. The superpositions with the anchor reveal the relative phases that force consistency.

The mixed-state transformation sends ρ to $\mathcal{V}\rho\mathcal{V}^{-1}$. Each image is a complex-linear operator on $\mathcal{H}$, even when $\mathcal{V}$ is antilinear, because the two antilinear factors cancel. The transformation preserves convex combinations; on the full complex operator space it is conjugate-linear when $\mathcal{V}$ is antiunitary. Its spectral form is $\sum_j p_j \left|\mathcal{V}\varphi_j\right\rangle\!\left\langle\mathcal{V}\varphi_j\right|$; the eigenvalues and trace are unchanged. A related theorem of Kadison starts with the convex structure of mixed states rather than transition probabilities (named only; Moretti, Thm 12.14, p. 682).

Self-check 4.2

1. Why is the anchor coefficient nonzero in the construction of W?
2. Which equality proves real linearity before complex linearity is known?
3. Where is dimension at least two needed for uniqueness of the type?

4.3 Time reversal includes domains

On $L^2(\mathbb{R})$ let $(\Theta\psi)(x) = \overline{\psi(x)}$. Conjugation is an onto antilinear isometry, so it is antiunitary and $\Theta^2 = \mathbb{1}$. The position domain $D(Q) = \{\psi : x\psi \in L^2\}$ is invariant under it, and direct multiplication gives $\Theta Q\Theta^{-1} = Q$ with that domain.

For momentum use QM-II's Fourier characterisation, $P = \mathcal{F}^{-1}p\mathcal{F}$ on $\{\psi : p\mathcal{F}\psi \in L^2\}$. On Schwartz functions, the Fourier integral gives

$$\mathcal{F}(\Theta\psi)(p) = \overline{\mathcal{F}\psi(-p)}.$$

Both sides extend continuously to L^2 by S1 and S4. Reflection and conjugation preserve the condition $p\mathcal{F}\psi \in L^2$ and reverse the momentum multiplier's sign. Thus

$$\Theta D(P) = D(P), \qquad \Theta P\Theta^{-1} = -P.$$

This is an equality of operators with domains, not a derivative rule applied to arbitrary square-integrable functions. Positions are even and momenta odd under this time reversal.

If a self-adjoint H has a conjugation-invariant domain and $\Theta H\Theta^{-1} = H$, then $\Theta U_t\Theta^{-1} = U_{-t}$. Indeed the transformed group has Hamiltonian $-H$, equivalently generator $-H/\hbar$: antilinearity reverses the scalar i in its strong derivative on $D(H)$. QM-II, Proposition 7.9, on uniqueness of the unitary group with a given generator supplies the equality.

Self-check 4.3

1. What happens to the momentum-space domain condition under $p \mapsto -p$?
2. Why does antiunitarity reverse the sign of iH rather than simply commute through it?

4.4 A ring's persistent current

Recall the twisted momentum of QM-II, Theorem 3.10 (self-adjointness sketched there): $P_\theta = -i\hbar\, d/dx$ on absolutely continuous functions with $\psi' \in L^2(0,L)$ and $\psi(L) = e^{i\theta}\psi(0)$. We allow real θ, with equivalent twists modulo 2π. Its normalised eigenvectors and eigenvalues are

$$e_{\theta,k}(x) = L^{-1/2} e^{i(2\pi k+\theta)x/L}, \qquad p_k(\theta) = \frac{\hbar}{L}(2\pi k + \theta), \quad k \in \mathbb{Z}.$$

The same conjugation Θ on $L^2(0,L)$ takes the boundary condition at θ into the one at $-\theta$. On these domains its derivative changes the sign of momentum:

$$\Theta D(P_\theta) = D(P_{-\theta}), \qquad \Theta P_\theta \Theta^{-1} = -P_{-\theta}, \qquad \Theta e_{\theta,k} = e_{-\theta,-k}.$$

For $H_\theta = P_\theta^2/(2m)$ the spectral levels are

$$E_k(\theta) = \frac{\hbar^2}{2mL^2}(2\pi k + \theta)^2, \qquad E_k(-\theta) = E_{-k}(\theta).$$

Thus $\Theta H_\theta \Theta^{-1} = H_{-\theta}$ with domains. If $H_\theta\psi = E\psi$, then $H_{-\theta}\Theta\psi = E\Theta\psi$, since E is real. The paired Hamiltonians have identical eigenvalues and multiplicities, so their ground energies are even in flux. Wherever the ground branch is differentiable, its flux derivative is odd. The square's domain is also carried to the opposite twist's domain, since both successive applications of momentum must exist. Equivalently, its eigenbasis domain is the square-summability condition with fourth powers of $2\pi k + \theta$, invariant under $(k,\theta) \mapsto (-k,-\theta)$.

To fix the physical sign, let the particle have charge $q \neq 0$ and define the twist by $\theta = -q\Phi/\hbar$, where Φ is the magnetic flux. As in QM-II, §3.4, this follows by removing the constant vector potential Φ/L from the periodic Hamiltonian $(P_0 - q\Phi/L)^2/(2m)$. Write the periodic wavefunction as $\psi_{\mathrm{per}}(x) = e^{iq\Phi x/(\hbar L)}\varphi(x)$. Differentiation gives

$$(P_0 - q\Phi/L)\psi_{\mathrm{per}} = e^{iq\Phi x/(\hbar L)}(-i\hbar\varphi'), \qquad \varphi(L) = e^{-iq\Phi/\hbar}\varphi(0).$$

The phase transformation is unitary and preserves absolute continuity and square-integrable derivatives. It moves the flux dependence from the differential rule into the boundary condition, with exactly the declared sign. Squaring the transformed momentum gives the same level formula and transformed square domain. Define the level current as charge times velocity divided by circumference, $I_k = qp_k(\theta)/(mL)$. It equals the flux derivative:

$$I_k = -\frac{\partial E_k}{\partial \Phi} = \frac{q}{\hbar}\frac{\partial E_k}{\partial \theta} = \frac{q\hbar}{mL^2}(2\pi k + \theta).$$

Differentiating with respect to a dimensionless twist without its flux conversion would miss this factor and could reverse the quoted sign.

For $-\pi < \theta < \pi$ the unique ground branch is $k = 0$, so

$$E_g(\Phi) = \frac{q^2\Phi^2}{2mL^2}, \qquad I_g(\Phi) = -\frac{q^2\Phi}{mL^2} \quad (|q\Phi/\hbar| < \pi).$$

The current is odd in flux and vanishes at zero flux. The spectrum and ground energy are periodic with flux period $2\pi\hbar/|q|$, by relabelling k. At odd multiples of π in twist, two

lowest branches cross; the ground energy has a cusp. There is no single derivative or unique ground-state current there without specifying the state in the degenerate space. Time reversal pairs the opposite currents rather than selecting one.

The level crossings and their current branches are displayed in Figure 4.1. The plotted current ratio rises with twist on each branch; converting twist to flux gives the sign in the preceding formula.

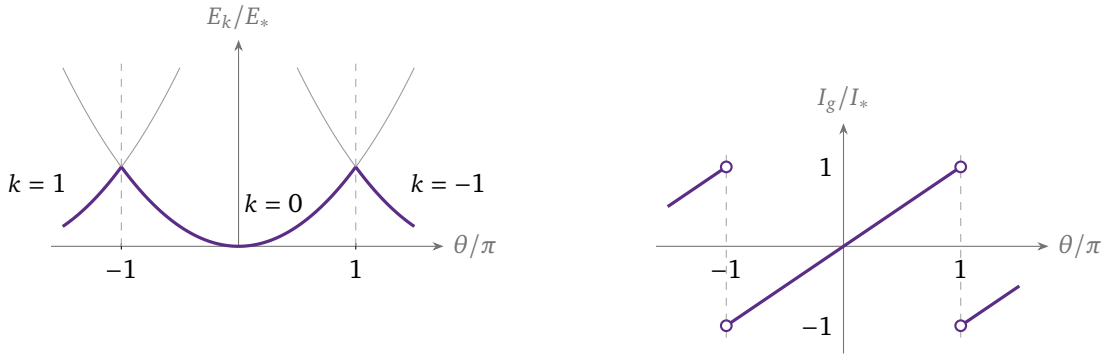


Figure 4.1: Ring levels and the ground-branch current, with $E_* = \hbar^2\pi^2/(2mL^2)$ and $I_* = q\hbar\pi/(mL^2)$, for $q \neq 0$. The thick energy curve is the lower envelope of the level parabolas; open current endpoints mark the two possible branch values at a crossing, not a uniquely selected current. The horizontal variable is twist, with $\theta = -q\Phi/\hbar$.

The physics

Persistent currents were measured through the magnetic response of mesoscopic metallic rings: for example, Lévy et al., *Phys. Rev. Lett.* **64**, 2074 (1990), and Chandrasekhar et al., **67**, 3578 (1991). Our calculation is one spinless particle on a clean ring, not a quantitative model of a disordered many-electron sample. It isolates the symmetry prediction: reversing flux reverses the current on corresponding nondegenerate branches. An amplitude prediction needs populations, disorder and interactions.

Self-check 4.4

1. Why is $\Theta P_\theta \Theta^{-1} = -P_\theta$ generally false at a fixed twist?
2. Which ground-state assumption fails at $\theta = \pi$?
3. With $\theta = -q\Phi/\hbar$, verify the sign in $I_k = -\partial E_k/\partial\Phi$.

4.5 A group in time cannot contain an antiunitary

On $\dim\mathcal{H} \geq 2$, suppose ray symmetries satisfy $\mathcal{S}_{s+t} = \mathcal{S}_s\mathcal{S}_t$ for all real s, t. Each is implemented by Wigner's theorem, but

$$\mathcal{S}_t = (\mathcal{S}_{t/2})^2.$$

The square of a unitary is linear; the square of an antiunitary is also linear, because conjugation of scalar coefficients occurs twice. Both squares are unitary. Type uniqueness in Theorem 4.3 therefore forces every $\mathcal{S}_t$ to have unitary type. Continuity has not even been used in this argument; it will be needed for the generator in L5.

On a qubit, conjugation in the σ_z basis sends ρ to $\bar{\rho} = \rho^{\mathsf{T}}$ and its Bloch vector to $(r_x, -r_y, r_z)$. It preserves transition probabilities, but it has antiunitary type. It cannot occur as one member of such a real one-parameter group. This does not make time reversal unphysical: time reversal changes the orientation of time, whereas the group of time translations moves forward or backward within that orientation.

Integer time labels would be different: set the ray map to the identity at every even integer and to conjugation at every odd integer. Since conjugation squares to the identity, these maps form a group over $\mathbb{Z}$ and include an antiunitary type. Extending it to all real times would require a half-step whose square had that type, which Wigner forbids. The conclusion is about a group with arbitrarily divisible time labels, not a prohibition of every discrete antilinear transformation.

The remaining obstruction is phase. Choosing a unitary implementation separately at every time need not give $U_{s+t} = U_s U_t$; it gives that identity only up to a phase. Wigner alone has not derived the evolution postulate. Lecture 5 states the additional continuity premise and the lifting theorem that remove this obstruction.

For a concrete example of uncoordinated phases, take any genuine group U_t and a fixed time scale $\tau > 0$. The representatives $\mathcal{V}_t = e^{i(t/\tau)^2} U_t$ induce exactly the same ray maps, but

$$\mathcal{V}_s \mathcal{V}_t = e^{-2ist/\tau^2} \mathcal{V}_{s+t}.$$

This follows by subtracting $(s+t)^2$ from $s^2 + t^2$. Each representative is unitary and depends continuously on time, yet the chosen family is not a group. Multiplying back by $e^{-i(t/\tau)^2}$ restores the group. Coordination of phases is therefore an extra step; continuity of an arbitrary choice of representatives does not by itself supply the group law.

Self-check 4.5

1. Why does the group argument use all real times rather than merely the integer powers of one map?

Which changes of pure states are symmetries?

For dimension at least two, a bijection of all rays preserving transition probabilities has one unitary or antiunitary implementation, unique up to phase and with a unique type. A real one-parameter group contains only unitary-type maps, but its individual phases still have to be coordinated. Time reversal is antiunitary and reverses momenta and magnetic flux; its domain action is part of each operator statement.

Summary. Wigner's theorem derives the allowed linear and antilinear forms from ray probabilities. The anchor proof establishes one global type, even in infinite dimension. On a ring, time reversal pairs opposite twists, levels and currents; the current derivative needs a nondegenerate branch. The square argument excludes antiunitaries from a real translation group. The group law up to phase remains to be addressed.

Exercises. Try Sheet 4 in the separate *Exercise sheets* booklet (English).

Self-check answers.

4.1 1. $\overline{i\psi} = -i\bar{\psi}$. 2. No; preserving the dynamics or commuting with its Hamiltonian is an additional condition.

4.2 1. Its transition with the anchor ray is $1/(1+\|z\|^2) > 0$. 2. Preservation of every real Gram entry, followed by the zero norm of the additivity error. 3. Two orthogonal vectors distinguish $[e_1 + ie_2]$ from $[e_1 - ie_2]$.

4.3 1. $\int p^2 |\mathcal{F}\psi(-p)|^2 dp$ equals the original weighted norm. 2. Antilinearity conjugates scalar coefficients, so i becomes $-i$.

4.4 1. The boundary condition changes to $-\theta$. 2. Uniqueness and differentiability of the lowest branch. 3. $\partial\theta/\partial\Phi = -q/\hbar$, giving the displayed plus factor $q/\hbar$ times the twist derivative.

4.5 The half-time map must exist; an integer-indexed group could have antiunitary-type maps at odd steps.

Sources. Galindo–Pascual, *Quantum Mechanics I*, §7.2 pp. 262–266; Weinberg, *The Quantum Theory of Fields I*, App. 2.A pp. 91–96; Moretti, *Spectral Theory and Quantum Mechanics*, Thm 12.11 pp. 675–681. The real-linearity/global-sign completion of the anchor proof is supplied here. The ring operator and its eigenbasis are recalled from QM-II, Theorem 3.10; the line operators from Theorem 4.6.